



GTPI302 Guided Learning My Learning Contract

to be included in E-Portfolio
(revised on Oct 2025)

Name of Student:	Nazrin Ally	Admin No.:	2402459H									
Name of Learning Facilitator:	Ms Vera Tay											
Project Title:	2D Rogue-like casual game development											
Brief Description of your project	Code a prototype/completed 2D wave defence game using the Unity game engine where the player must fight through progressively harder waves while upgrading their character to best fit their playstyle. Each playthrough will be different in terms of enemies, upgrades and waves ensuring each run is unique and ensures replayability.											
My Learning Goals	Refine Learning Goals as needed: <i>E.g. to work on my marketing skills.</i> - Develop intermediate-to-advanced Unity skills – learn to implement gameplay mechanics such as enemy AI, health/damage systems, and wave progression to build a functional and balanced 2D rogue-like game. - Create a professional e-portfolio to showcase my project process, reflections, and final prototype using a structured design - Work on my time-management											
	My project will help me cultivate these TP Student Profile attributes (choose three attributes) * Refer Appendix I for each TSP description <table border="0"> <tr> <td>☐ Inquisitive</td> <td>☐ Integrity</td> <td>☐ Critical thinking</td> </tr> <tr> <td>☒ Resilient</td> <td>☐ Empathy</td> <td>☐ Digitally Fluent</td> </tr> <tr> <td>☒ Growth-Oriented</td> <td>☒ Communicates Effectively</td> <td>☐ Innovative & Entrepreneurial</td> </tr> </table>			☐ Inquisitive	☐ Integrity	☐ Critical thinking	☒ Resilient	☐ Empathy	☐ Digitally Fluent	☒ Growth-Oriented	☒ Communicates Effectively	☐ Innovative & Entrepreneurial
☐ Inquisitive	☐ Integrity	☐ Critical thinking										
☒ Resilient	☐ Empathy	☐ Digitally Fluent										
☒ Growth-Oriented	☒ Communicates Effectively	☐ Innovative & Entrepreneurial										
	For each chosen attribute, how would you rate yourself on a scale of 1-5 at this point (e.g. in the context of school work, personal life, part-time jobs)? 1: Never used or exhibited this attribute before 2: Rarely using or exhibiting this attribute 3: Sometime using or exhibiting this attribute 4: Most of the time using or exhibiting this attribute 5: Always using or exhibiting this attribute											
	For each chosen attribute, provide your current rating and explain how you can develop each attribute through GL. Attribute 1: Resilient 4: Coding a full game will be challenging. I will develop resilience by debugging issues patiently, adapting to feedback, and persisting through design and coding setbacks. Attribute 2: Growth 3: I will expand my technical and creative skills by learning new Unity features and improving my collaboration and reflection through this project.. Attribute 3: Communicate Effectively 4: Working in a team requires regular discussion, feedback sharing, and documentation. I will improve by practising clear, impactful, and											

	consistent communication with my partner and facilitator.		
Target beneficiary/ beneficiaries (if applicable):	N/A		
Plan your learning; how do you achieve your own learning goals?	What new knowledge & skills I already know Movement functions Character animation sequence Good coding practices E-portfolio		
	What new knowledge & skills will I need to complete this project? New knowledge needed: Health and Damage system Enemy AI (movement and attack pattern) Enemy coin system (how much coins player gets per kill) New Skills needed: Code for movement and tracking of enemy Animation sequence for each enemy, like their attacks, death anim, and hurt anim Incremental Health and Damage according to each wave as it gets harder		
	What possible challenges would I need to overcome for my learning project? Time management, I have to balance schoolwork and GL project together. As im from the AI course, most of my projects will be related to coding as well and sometimes I will take too long to debug my school projects leaving me no time to debug the problems found in the code for my GL project. Lack of knowledge to complete certain aspects of my game. Since im a beginner at using Unity, most of my skills were self-taught, therefore they would be patches in my knowledge, luckily my partner is competent at Unity and therefore we can help each other out.		
	I plan to get help from: Please tick and name the specific organisation(s)/expert(s) you would consult, include designation, names, telephone numbers and email ☐ Partner departments in TP: _____ ☐ Partner Schools in TP: _____ ☐ External Partner Organisation: _____ ☐ External Resource People: _____ ☒ Others: YouTube		
	I plan to use these learning resources: Please tick and name the resources you would use. ☐ Books/newspapers: _____ ☒ Online learning/courses: <u>Online Unity Courses or other free courses</u> ☐ Internet journals/articles: _____ ☐ Experiential learning: _____ ☒ Documentaries/videos: <u>YouTube tutorials, online forums</u> ☐ Others (e.g. seminars, interviews, performances, lectures, workshops) _____		
Action Plan Timelines & Deliverables	Week	Actions	Deliverables
	Plan for the entire semester on a weekly basis Week 1	Specify the actions you will be taking to complete your project throughout the semester <i>Contact my facilitator and discuss the action plan and other deliverables</i>	Specify concrete and tangible deliverables you will produce <i>Start on Learning Log, Game Plan and Learning Contract</i>
	Week 2	<i>Continue with self-taught Unity Learning, set up environment and</i>	<i>Set up Unity environment, continue with online materials youtube vids, free courses etc</i>

		<i>game design, start bi-weekly reflection</i>	
	Week 3	<i>Work on the Unity game, update my partner accordingly with our designated task e.g: enemy movement prototype first attempt</i>	<i>Update game design document on progression, meet up with partner</i>
	Week 4 (Flex Week 1)	<i>Continue progress for enemy movements, learn from mistakes and improve from week 3s first test</i>	<i>Check in with progress from partner and continue game prototype, start on learning log 2</i>
	Week 5	<i>Contact facilitator to show progress and get consult plus feedback</i>	<i>Submit learning log 2</i>
	Week 6	<i>Find 3 play testers and have them do the google form for future improvement</i>	<i>Google Form, Meet up with partner</i>
	Week 7	<i>Polish enemy movement and attack patterns, start on more and begin e portfolio</i>	<i>E portfolio, Game prototype</i>
	Week 8	<i>Arrange third consultation for progress and feedback</i>	<i>Start and submit learning log 3, Meet up with partner</i>
	Week 9 (Term Test Week)	<i>Review google form feedback and discuss about next addition</i>	<i>Game prototype</i>
	Week 10 (Term Break)	<i>Continue working on next enemy, leaving UI and health system to partner</i>	<i>Game prototype, Meet up with partner</i>
	Week 11 (Term Break)	<i>Polish enemy movement and attack, work on e portfolio</i>	<i>Game Prototype, e portfolio</i>
	Week 12	<i>Communicate with group mate on progress and start learning log 4</i>	<i>Learning log 4, Meet up with partner</i>
	Week 13	<i>Finish e portfolio, reflection and polish the game (handling any small errors or bugs)</i>	<i>Completed e-Portfolio and game completed</i>
	Week 14	<i>Arrange fourth consultation to update on game progress and feedback of e portfolio</i>	<i>E portfolio and learning log, Meet up with partner</i>
	Week 15 (Flex Week 2)	<i>Finishinh up final touches, practicing oral interview</i>	<i>nil</i>
	Week 16	<i>Oral reflection interview</i>	<i>Final meet up with partner</i>
	Week 17	<i>Finish e portfolio</i>	<i>E portfolio submission</i>
	Please customise if your project commences prior to the regular semester timeframe.		
Plan your E-portfolio;	What do you include in your E-portfolio? <i>e..g. Report, prototype, checklist, glossary of terms, reflection, draft designs, surveys, survey analyses, interview transcripts, etc. (50-100 words).</i>		
Learning evidence for E-portfolio	Working prototype of the game, video demo, timeline with progress updates and challenges faced, learning contract and log, survey,interview and their analysis, reflections, brief description of our game and myself, why we chose to do this		

Annex 1

TP Student Profile attributes

Lifelong Learner

TSP	TSP Attributes
-----	----------------

Inquisitive	Being curious, explore, question, and seek knowledge across disciplines.
Resilient	Adapt, persevere through challenges, and thrive in changing environments.
Growth-oriented	Seek continuous improvement and take ownership in learning.

Values-centered Leader

TSP	TSP Attributes
Integrity	Act with ethics, honesty and strong moral principles.
Empathy	Understand and respect diverse perspectives with compassion and global awareness.
Communicates Effectively	Express ideas clearly and engage meaningfully with others.

Future-oriented Creator

TSP	TSP Attributes
Critical Thinking	Analyse information objectively to make informed decisions.
Digitally Fluent	Navigate digital tools, technologies, and information discerningly.
Innovative & Entrepreneurial	Create new ideas, take initiatives, take calculated risks, and embrace opportunities.